Quinnipiac
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Evaluating Physics-Informed Learning for Fluid Dynamics

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Does training a neural network on the laws of physics make its answers more trustworthy, and is it worth the extra computing cost?

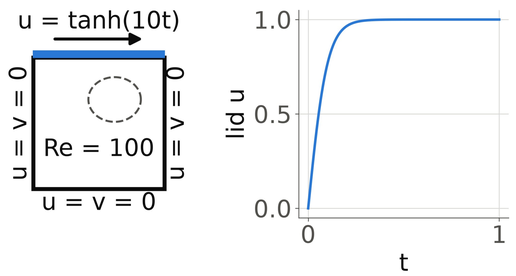
Why it matters

Fluid simulations are accurate but slow. A trained neural network answers in milliseconds, but nothing forces its answer to obey physics.

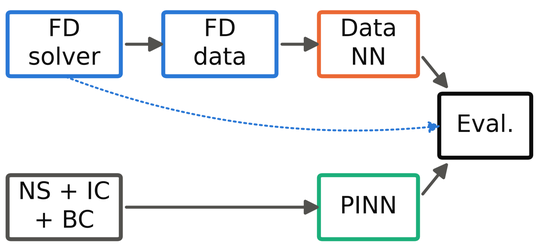
A physics-informed neural network (PINN) learns from the flow equations themselves instead of from simulation data.

How we tested it

A lid slides across a box of fluid and sets it spinning (lid-driven cavity, Reynolds number Re = 100).



Three ways to get the same flow, scored by one evaluator:

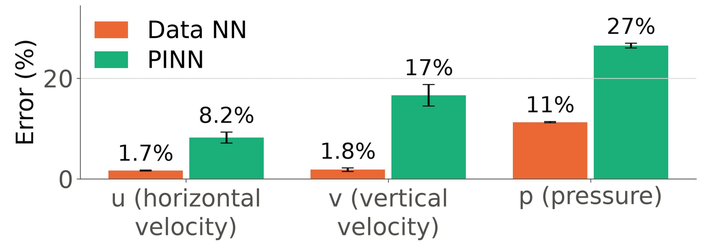


- Same network size for both (17,091 parameters), 5 training seeds each.
- Scored on points and whole time steps never used in training.
- Simulation checked against two published studies: vortex center within 0.004 of Marchi et al. (less than one grid cell).

What we found (Re = 100)

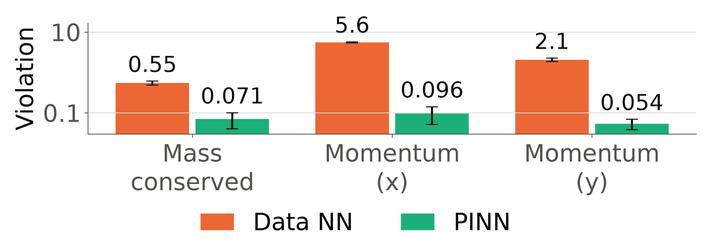
1 Learning from data gives the closest answer.

Error in u: 1.7% for the data NN, 8.2% for the PINN (mean of 5 seeds).



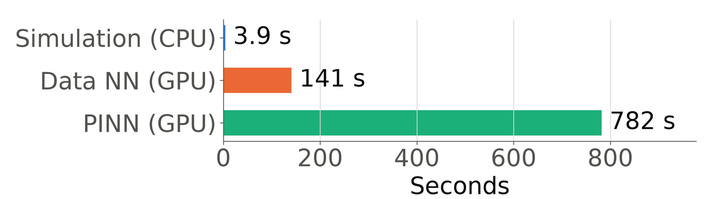
2 The PINN's answers obey physics better.

How far each flow breaks the equations of motion (lower is better).



3 Physics costs computing time.

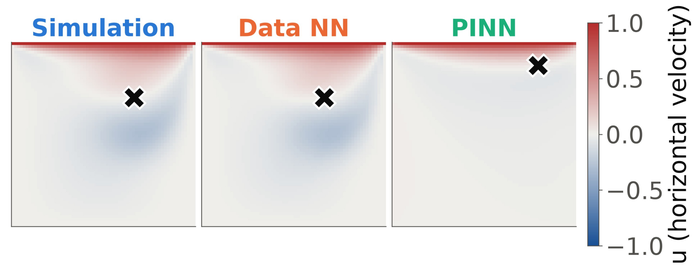
Time to a usable flow (data NN time includes making its simulation data). Once trained, either network returns the whole flow in about 10 ms.



Where it breaks

4 A small physics error is not a correct answer.

At a faster flow (Re = 400) the PINN broke conservation of mass less than the data NN (0.03 vs 0.55), yet its vortex (✕) sat by the lid. Error in u: 36% (PINN, all 5 seeds) vs 2.3% (data NN).



5 Does more training help?

Runs trained 5× longer (100,000 steps) are in progress.

What the numbers show

- Training on physics made the network obey physics, not match the simulation better.
- It cost more: about 782 s vs 137 s of GPU training.
- A network can satisfy the equations and still be wrong, so it must be checked against a trusted reference.

[1] Marchi, C. H., Suero, R., and Araki, L. K. 2009. The lid-driven square cavity flow: numerical solution with a 1024 x 1024 grid. J. Braz. Soc. Mech. Sci. Eng. 31, 3, 186–198.
[2] Hou, S., Zou, Q., Chen, S., Doolen, G., and Cogley, A. C. 1995. Simulation of cavity flow by the lattice Boltzmann method. J. Comput. Phys. 118, 2, 329–347.
[3] Raissi, M., Perdikaris, P., and Karniadakis, G. E. 2019. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. J. Comput. Phys. 378, 686–707.

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